

Technology data of Nylon film (BOPA film)

Biaxially Oriented Nylon Film

	Units	TPA 15	TPA 25	Standard / Test Method
Thickness	μm	15 ± 3%	25 ± 3%	DIN 53370
Yield	m ² /kg	58 ± 3%	34.8 ± 3%	Calculated
Melting Point	°C	220 ± 2	220 ± 2	DSC Machine
Dimensional Stability	% (MD / TD)	< 2.5 / < 1.5	< 2.5 / < 1.5	160°C 5 min (ABI)
Tensile Strength	MPa	> 220	> 220	ASTM D 882
Modulus of Elasticity	MPa	< 1315	< 1315	ASTM D 882
Elongation at Break	% (MD / TD)	120 / 100 ± 30	120 / 100 ± 30	ASTM D 882
COF (film to film)	(static / dynamic)	< 0.8 / < 0.8	< 0.8 / < 0.8	ASTM D 1894
COF (film to metal)	(static / dynamic)	< 0.4 / < 0.4	< 0.4 / < 0.4	ASTM D 1894
Corona Treatment	mJ / m	Min 52	Min 52	ASTM D 2578
Haze	%	< 3	< 3	ASTM D 1003
Gloss	%	> 80	> 80	ASTM D 2457
Oxygen Transmission Rate (OTR)	cm ³ / m ² / 24hrs	< 30	< 20	23°C, 50% RH; ASTM D 3985
Water Vapor Transmission Rate (WVTR)	g / m ² / 24hrs	< 180	< 126	38°C, 90% RH

TPA film. complies with FDA regulation 21 CFR, Part 177, Subpart B, Paragraph 177.1500,
is approved for meat, cheese, seafood and liquid packing by the Canadian Food Inspection Agency,
complies with European Directive 2002/72/EC relating to materials in direct contact with food.
Recommendation and data given are based on experience to date. No liability can be assumed in connection with their usage.

10/1/2014

Typical Film Properties 典型的薄膜特性

BOPA – Film, 15 μ m, 3-Layer BOPA薄膜 – 15微米, 3层

No	Description	Unit	Average	Method
1	Thickness	-TD μ m	14,8	DIN 53370
2	Tensile Strength 拉伸强度	-MD N/mm^2	270	DIN 53455
		TD N/mm^2	336	DIN 53455
3	Elongation at Break 断裂伸长率	%	120	DIN 53455
		-TD %	90	DIN 53455
4	Modulus of Elasticity 弹性模量	-MD N/mm^2	2910	DIN 53457
		-TD N/mm^2	3250	DIN 53457
5	Coefficient of Friction 摩擦系数	F/F- T/O - μ s	0,47	DIN 53375
		T/O - μ k	0,43	DIN 53375
6	Haze	%	1,75	ASTM 1003
7	Gloss	—	160	ASTM 2457
8	Treatment Level	—	56	ASTM 2578
9	Thermal Shrinkage (160°C, 5 min) 热缩	-MD %	1,5	BMS TT 02
		-TD %	0,8	BMS TT 02
10	Oxygen Transmission (23°C, 50%RH) 透氧	$cm^3/m^2 dbar$	30	ASTM D 3985
11	Water Vapour Transmission (38°C, 90%RH) 透水蒸汽	$g/m^2 d$	210	ASTM F-1249

双向拉伸尼龙薄膜
BOPA FILM

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A		B		C		D		E	
PROPERTY		METRIC(S.I.) UNITS				TEST METHOD			
		UNIT		TYPICAL VALUE					
THICKNESS		μm		15					
YIELD		mm ² / kg		57.5					
TENSILE	MD	kg / mm ²		33		ASTM			
STRENGTH	TD			32		D-882			
ELONGATION	MD	%		151		ASTM			
AT BREAK	TD			150		D-882			
FLEX	MD	kg / mm ²		5.8		ASTM			
	TD			5.6		D-882			
MODULUS	MD	kg / mm ²		326		ASTM			
	TD			3.4		D-882			
HEAT-SEAL	MD	%		2.2		ASTM			
	TD			0.4		D-2305			
PERMEATION	MD	μm		0.4		ASTM			
	TD			2.6		D-1894			
HAZE				2.6		D-1003			
WETTING	IN	dyne/cm		52		ASTM			
TENSION	OUT					D-2578			
WVTR		g/m ² *day		260		JIS Z-0208			
OXYGEN PERMEABILIT		cc/m ² *day		52.6		ASTM D-3985			

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BOPA FILM (Test report)

Typical physical properties of products:

ITEM	TEST METHOD	UNIT	TYPICAL VALUE
Thickness	GB6672	μm	15
Density	GB1033	g/cm^3	1.15
Tensile strength (MD/TD)	GB13022	MPa	240/255
Shrinkage rate (MD/TD)	GB12026 (160℃X5min)	%	1.5/1.0
Surface Tension	GB/T14216	mN/m	54
Haze	GB2410	%	3.0
Oxygen permeability (24hr)	GB1038	$\text{cm}^3/\text{m}^2 \cdot \text{l atm}$	30